Shamir Khan

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Education

Bachelor of Science in Computer Science — **Expected April 2028**
Wilfrid Laurier University — *Waterloo, ON*

Relevant Coursework: Object-Oriented Programming (OOP), Data Structures & Algorithms, Distributed Systems, Operating Systems, Database Systems, Computer Networks, Systems Architecture, Discrete Mathematics

Experience

AI Software Engineer (Client Project) — **Oct 2026 – Present**
Go Fig Bookz (via Riipen Labs) — *Remote, ON*

- **Architected an autonomous AI bookkeeping agent** using **TypeScript** and **Node.js**, programmatically reconciling multi-bank statements with **100% tie-out** and **$0.00 discrepancy** across $17K+ in client disbursements.
- **Engineered three-way document matching algorithms** and automated HST tax calculation workflows, detecting duplicate vendor invoices and reducing manual accounting reconciliation effort by **80%**.
- **Authored comprehensive Vitest test suites** covering 43 automated tests across 9 suites, validating end-to-end multi-bank statement parsing, deterministic ledger updates, and zero-regression audit pipelines.

Lead Web Developer & Systems Administrator — **Jan 2026 – Present**
Wilfrid Laurier University Student Association (WLU MSA) — *Waterloo, ON*

- **Architected and deployed a production web platform** serving **250+ active campus members** using **Next.js**, **React**, **TypeScript**, and **Tailwind CSS**, maintaining **99.9% uptime** (wlumsa.org).
- **Engineered dynamic donation and event modules** integrated with **Payload CMS** and payment webhooks, supporting student-led initiatives that processed **$50K+** in charitable contributions.

Projects

CloudGuard – Cloud Security Posture Management (CSPM) | *Live Site* | *GitHub* — **July 2026**

- **Architected an event-driven distributed microservices platform** across 5 containerized services connected via **Apache Kafka**, achieving **99.9% uptime** and handling multi-cloud telemetry ingestion across AWS, Azure, and GCP.
- **Built a native C++ real-time telemetry agent** using Windows kernel APIs (`GlobalMemoryStatusEx`, `GetSystemTimes`) and POSIX parsers to stream machine metrics to Kafka topics at sub-millisecond overhead.
- **Developed an unsupervised scikit-learn IsolationForest ML pipeline** scoring live Kafka streams in **507ms (p50)**, detecting SSH brute-force intrusions and billing spikes to cut incident response MTTR by **85%**.

NexusQueue – Distributed Asynchronous Task & DAG Engine | *Live Site* | *GitHub* — **Nov 2026**

- **Engineered a distributed background task engine** benchmarked at **119,000 jobs/sec** enqueue throughput and **3,418 tasks/sec** execution across 50 concurrent worker pipelines with strict at-least-once delivery guarantees.
- **Implemented a Binary Min-Heap Priority Queue** with $O(\log n)$ scheduling and FIFO tie-breaking, combined with distributed **Token Bucket** rate limiters and AWS Full Jitter backoff against thundering-herd storms.
- **Architected a DAG Workflow Scheduler** resolving task dependency trees via **Kahn's topological sort**, paired with an automated **Zombie Worker Watchdog** and live WebSocket telemetry dashboard.

LedgerMesh – High-Concurrency Double-Entry Ledger Core | *Live Site* | *GitHub* — **Nov 2026**

- **Architected an institutional clearing engine** benchmarked at **10,790 tx/sec** with a median ($p50$) latency of **0.05ms**, enforcing mathematical double-entry invariants (**Total Debits = Total Credits**) and zero floating-point drift.
- **Designed Optimistic Concurrency Control (OCC)** with account versioning and backoff, eliminating race conditions across 20 concurrent worker coroutines, paired with a cryptographic **SHA-256 Merkle audit trail**.

Technical Skills Summary

Languages: C++, Python, TypeScript, Java, JavaScript, Scala, Kotlin, SQL, HTML5/CSS3, C
Distributed Systems & Cloud: Apache Kafka, Apache Spark, Redis Streams, Docker, WebSockets, GitHub Actions (CI/CD), Linux/Unix, Render, Vercel, AWS
Frameworks & Databases: FastAPI, Node.js, Express, React, Next.js, Spring Boot, PostgreSQL, MongoDB, SQLite, NumPy, Pandas, OpenCV, Tailwind CSS
Core Concepts: Optimistic Concurrency Control (OCC), Binary Min-Heaps, DAG Scheduling (Kahn's Algorithm), SHA-256 Merkle Trees, Token Bucket Rate Limiting, Full Jitter Backoff, Object-Oriented Programming (OOP), Test-Driven Development (TDD), RESTful APIs